

High tofu intake is associated with worse memory in elderly Indonesian men and women.

BACKGROUND/AIMS: Cell culture studies suggest that phytoestrogens, abundant in soy products such as tempe and tofu, could protect against cognitive decline. Paradoxically, the Honolulu Asia Aging Study reported an increased risk for cognitive impairment and other dementia markers with high tofu (soybean curd) intake. **METHODS:** A cross-sectional study was carried out in 2 rural sites (Borobudur and Sumedang) and 1 urban site (Jakarta) among mainly Javanese and Sundanese elderly ($n = 719$, 52-98 years of age). Memory was measured using a word learning test sensitive to dementia and soy consumption was assessed using Food Frequency Questionnaire items. **RESULTS:** High tofu consumption was associated with worse memory ($\beta = -0.18$, $p < 0.01$, 95% CI = -0.34 to -0.06), while high tempe consumption (a fermented whole soybean product) was independently related to better memory ($\beta = 0.12$, $p < 0.05$, 95% CI = 0.00-0.28), particularly in participants over 68 years of age. Fruit consumption also had an independent positive association. The analyses were controlled for age, sex, education, site and intake of other foods. **CONCLUSION:** The results for tofu consumption as a risk factor for low memory function may tie in with the Honolulu Asia Aging Study data. It is unclear whether these negative associations could be attributed to potential toxins or to its phytoestrogen levels. Estrogen (through which receptors phytoestrogens can exert effects) was found to increase dementia risk in women over 65 years of age. Tempe contains high levels of phytoestrogens, but (due to fermentation) also exhibits high folate levels which may exert protective effects. Future studies should validate these findings and investigate potential mechanisms. Copyright 2008 S. Karger AG, Basel.